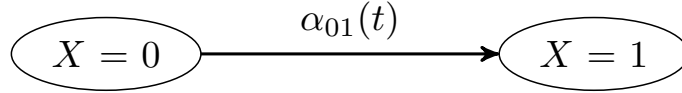


Continuous-Time Markov Chains

1.1 Overview

Suppose $X(t)$ is a continuous-time non-homogeneous Markov chain with a finite state space $\mathcal{S} = \{0, 1, \dots, J\}$. Consider first the two-state chain $\mathcal{S} = \{0, 1\}$ underlying standard survival analysis:



The *intensity* or *hazard matrix* is:

$$\boldsymbol{\alpha}(t) = \begin{pmatrix} -\alpha_{01}(t) & \alpha_{01}(t) \\ 0 & 0 \end{pmatrix}.$$

Entry $(1, 2)$ is $\alpha_{01}(t)$, the hazard of transitioning from state 0 to state 1. Entry $(1, 1)$ is the negation of entry $(1, 2)$, indicating that if $X(t)$ enters state 1, then $X(t)$ has left state 0. The second row of $\boldsymbol{\alpha}(t)$ is zero, indicating that transitions from state 1 back to state 0 are not allowed. The event time T is defined as $T = \inf\{t > 0 : X_t \neq 0\}$. The *transition matrix* is:

$$\mathbf{P}(t) = \begin{pmatrix} S(t) & 1 - S(t) \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} e^{-A_{01}(t)} & 1 - e^{-A_{01}(t)} \\ 0 & 1 \end{pmatrix}$$

where $A_{01}(t) = \int_0^t \alpha_{01}(s) ds$ is the cumulative hazard of transitioning $0 \rightarrow 1$ within $(0, t]$. More generally, for an inhomogeneous Markov chain, $\mathbf{P}(s, t)$ is the transition matrix with elements:

$$P_{ij}(s, t) = \mathbb{P}\{X(t) = j | X(s) = i\}.$$

The transition matrix is linked to the hazard matrix by:

$$\frac{\partial}{\partial t} \mathbf{P}(s, t) = \mathbf{P}(s, t) \boldsymbol{\alpha}(t), \quad (1.1)$$

where the hazard matrix is formally defined by:

$$\boldsymbol{\alpha}(t) = \lim_{\delta \downarrow 0} \frac{1}{\delta} \{ \mathbf{P}(t, t + \delta) - \mathbf{I} \}.$$

(1.1) is the *forward Kolmogorov equation*. The complementary integral equation is:

$$\mathbf{P}(s, t) = \mathbf{I} + \int_s^t \mathbf{P}(s, u-) d\mathbf{A}(u).$$

This expression, using $d\mathbf{A}(u)$ in place of $\boldsymbol{\alpha}(u)du$, remains valid even when the elements of the transition matrix are not differentiable. The solution to (1.1) takes the form of a matrix-valued *product integral*:

$$\mathbf{P}(s, t) = \prod_{s < u \leq t} \{\mathbf{I} + d\mathbf{A}(u)\}. \quad (1.2)$$

In the absolutely continuous case, $d\mathbf{A}(u) = \boldsymbol{\alpha}(u)du$, where $\boldsymbol{\alpha}(u)$ is the matrix of element-wise derivatives of the cumulative hazard matrix $\mathbf{A}(u)$. For the simple two-state model:

$$\mathbf{A}(t) = \begin{pmatrix} -\int_0^t \alpha_{01}(u)du & \int_0^t \alpha_{01}(u)du \\ 0 & 0 \end{pmatrix},$$

and:

$$\mathbf{P}(0, t) = \prod_{0 < u \leq t} \begin{pmatrix} 1 - \alpha_{01}(u)du & \alpha_{01}(u)du \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} e^{-A_{01}(t)} & 1 - e^{-A_{01}(t)} \\ 0 & 1 \end{pmatrix}.$$

Closed forms for the product integral equation (1.2) are typically only available for simple models, such as the competing risk or illness-death models.

1.2 Aalen-Johansen Estimator

Let $\tau_1 < \dots < \tau_K$ denote the distinct observation times. The transition matrix $\mathbf{P}(s, t)$ is consistently estimated via the **Aalen-Johansen** estimator:

$$\hat{\mathbf{P}}(s, t) = \prod_{s < \tau_j \leq t} \{\mathbf{I} + d\hat{\mathbf{A}}(\tau_j)\}. \quad (1.3)$$

1.2.1 Two-State Model

Example 1.1. Consider the two-state model. If τ_j is an event time:

$$\mathbf{I} + d\hat{\mathbf{A}}(\tau_j) = \begin{pmatrix} 1 - \frac{dN(\tau_j)}{Y(\tau_j)} & \frac{dN(\tau_j)}{Y(\tau_j)} \\ 0 & 1 \end{pmatrix}.$$

Let $h_j = dN_j/Y_j$. The product of the first two Aalen-Johansen matrices is:

$$\begin{pmatrix} 1 - h_1 & h_1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 - h_2 & h_2 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} (1 - h_1)(1 - h_2) & (1 - h_1)h_2 + h_1 \\ 0 & 1 \end{pmatrix}$$

The product of the first three matrices will be:

$$\begin{pmatrix} (1 - h_1)(1 - h_2)(1 - h_3) & (1 - h_1)(1 - h_2)h_3 + (1 - h_1)h_2 + h_1 \\ 0 & 1 \end{pmatrix}$$

And in general, the product over the first k event times is:

$$\hat{P}_k = \prod_{i=1}^k \{I + d\hat{A}_i\} = \begin{pmatrix} \hat{S}_k & \hat{F}_k \\ 0 & 1 \end{pmatrix},$$

where $\hat{S}_k$ is the standard Kaplan-Meier estimator:

$$\hat{S}_k = \prod_{i=1}^k (1 - h_i),$$

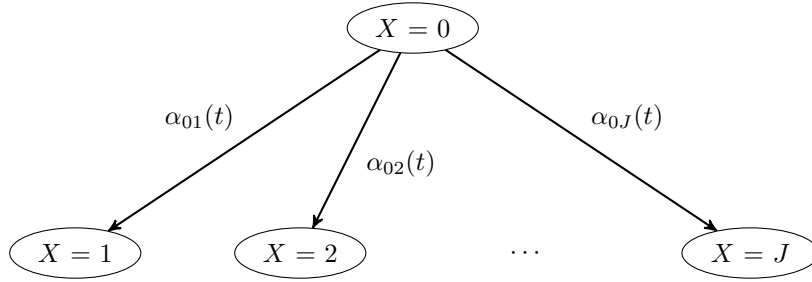
and $\hat{F}_k$ is a corresponding estimator of the cumulative incidence:

$$\hat{F}_k = \sum_{i=1}^k h_i \prod_{j < i} (1 - h_j).$$



1.2.2 Competing Risks Model

Example 1.2. Consider the J -state competing risk model:



Define $N_{0\bullet}(t) = \sum_{j=1}^J N_{0j}(t)$ as the total number of transitions out of state 0 by time t .

If τ_j is an event time:

$$I + d\hat{A}(\tau_j) = \begin{pmatrix} 1 - \frac{dN_{0\bullet}(\tau_j)}{Y_0(\tau_j)} & \frac{dN_{01}(\tau_j)}{Y_0(\tau_j)} & \frac{dN_{02}(\tau_j)}{Y_0(\tau_j)} & \dots & \frac{dN_{0J}(\tau_j)}{Y_0(\tau_j)} \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{pmatrix}$$

Matrix multiplications, as in the two-state model, show that:

$$\hat{P}_{00}(s, t) = \prod_{s < u \leq t} \left(1 - \frac{dN_{0\bullet}(u)}{Y_0(u)} \right),$$

which is the Kaplan-Meier estimator based on the total number of events.

For $j \in \{1, \dots, J\}$, the estimator of the cumulative incidence function is:

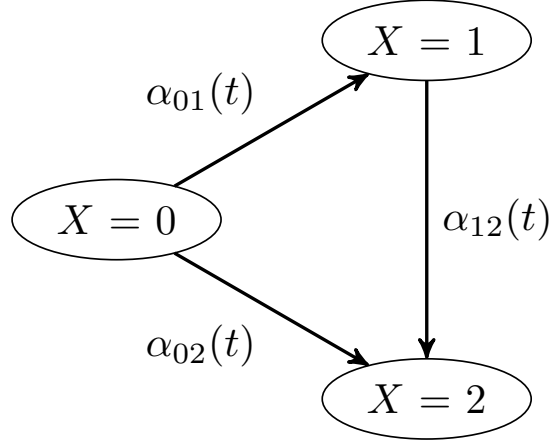
$$\hat{P}_{0j}(s, t) = \int_s^t \hat{P}_{00}(s, u-) \cdot \frac{dN_{0j}(u)}{Y_0(u)}.$$

Here $P_{0j}(s, t) = \mathbb{P}(s < T \leq t, X_T = j)$.



1.2.3 Illness-Death Model

Example 1.3. Consider the illness-death model without recovery:



Let $Y_0(t)$ and $Y_1(t)$ denote the number of individuals in states 0 and 1 just prior to time t . Let $N_{01}(t)$, $N_{02}(t)$, and $N_{12}(t)$ count the numbers of transitions between the corresponding states by time t .

If τ_j is a transition time:

$$\mathbf{I} + d\hat{\mathbf{A}}(\tau_j) = \begin{pmatrix} 1 - \frac{dN_{0\bullet}(\tau_j)}{Y_0(\tau_j)} & \frac{dN_{01}(\tau_j)}{Y_0(\tau_j)} & \frac{dN_{02}(\tau_j)}{Y_0(\tau_j)} \\ 0 & 1 - \frac{dN_{12}(\tau_j)}{Y_1(\tau_j)} & \frac{dN_{12}(\tau_j)}{Y_1(\tau_j)} \\ 0 & 0 & 1 \end{pmatrix}$$

Matrix multiplications show that the survival function for state 0 is:

$$\hat{P}_{00}(s, t) = \prod_{s < u \leq t} \left(1 - \frac{dN_{0\bullet}(u)}{Y_0(u)} \right),$$

which is analogous to the competing risk model.

The survival function for state 1 is:

$$\hat{P}_{11}(s, t) = \prod_{s < u \leq t} \left(1 - \frac{dN_{12}(u)}{Y_1(u)} \right),$$

which is analogous to the two-state model.

The probability of starting in state 0 at time s and occupying state 1 at time $t > s$ is:

$$\hat{P}_{01}(s, t) = \int_s^t \hat{P}_{00}(s, u-) \cdot \frac{dN_{01}(u)}{Y_0(u)} \cdot \hat{P}_{11}(u, t).$$

The limiting expressions for the transition probabilities are:

$$P_{00}(s, t) = \exp \left\{ - \int_s^t \alpha_{0\bullet}(u) du \right\}$$

where $\alpha_{0\bullet}(u) = \alpha_{01}(u) + \alpha_{02}(u)$;

$$P_{11}(s, t) = \exp \left\{ - \int_s^t \alpha_{12}(u) du \right\};$$

and:

$$P_{01}(s, t) = \int_s^t P_{00}(s, u-) \alpha_{01}(u) P_{11}(u, t) du.$$

Finally, note that $P_{02}(s, t) = 1 - P_{00}(s, t) - P_{01}(s, t)$. ♠

1.3 Inference

Suppose again that $X(t)$ is a continuous-time Markov chain with finite state space $\mathcal{S} = \{0, 1, \dots, J\}$. The asymptotic distribution of the Aalen-Johansen estimator (1.3) may be derived by vectorizing $\hat{\mathbf{P}}(s, t)$, then expressing the covariance of $\text{vec}(\hat{\mathbf{P}})$ as a matrix-valued mean zero martingale. The resulting estimator of the covariance between $\hat{P}_{ab}(s, t)$ and $\hat{P}_{cd}(s, t)$ is: $\hat{\mathbb{C}}\{\hat{P}_{ab}(s, t), \hat{P}_{cd}(s, t)\} =$

$$\sum_{i=0}^J \sum_{k \neq i} \int_s^t \hat{P}_{ak}(s, u) \hat{P}_{ck}(s, u) \{ \hat{P}_{ib}(u, t) - \hat{P}_{kb}(u, t) \} \{ \hat{P}_{id}(u, t) - \hat{P}_{kd}(u, t) \} \cdot \frac{dN_{ki}(u)}{Y_k^2(u)} \quad (1.4)$$

When $a = c$ and $b = d$, such that the variance is of interest:

$$\hat{\mathbb{V}}\{\hat{P}_{ab}(s, t)\} = \sum_{i=0}^J \sum_{k \neq i} \int_s^t \hat{P}_{ak}^2(s, u) \{ \hat{P}_{ib}(u, t) - \hat{P}_{kb}(u, t) \}^2 \cdot \frac{dN_{ki}(u)}{Y_k^2(u)}$$

See [1] (IV.4.1.3) for details.

1.3.1 Two-State Model

Example 1.4. Consider the two-state chain $\mathcal{S} = \{0, 1\}$. The Aalen-Johansen covariance estimator for $\hat{P}_{00}(0, t)$, which is the standard survival function. For this model, $\hat{P}_{10}(s, t) = 0$ and $dN_{10}(u) = 0$, such that only the $(i, k) = (1, 0)$ term contributes:

$$\begin{aligned} \hat{\mathbb{V}}\{\hat{P}_{00}(0, t)\} &= \int_0^t \hat{P}_{00}^2(0, u) \{0 - \hat{P}_{00}(u, t)\}^2 \cdot \frac{dN_{01}(u)}{Y_0^2(u)} \\ &= \int_0^t \hat{P}_{00}^2(0, u) \hat{P}_{00}^2(u, t) \cdot \frac{dN_{01}(u)}{Y_0^2(u)}. \end{aligned}$$

The squared term simplifies as:

$$\hat{P}_{00}^2(0, u) \hat{P}_{00}^2(u, t) = \{ \hat{P}_{00}(0, u) \hat{P}_{00}(u, t) \}^2 = \{ \hat{P}_{00}(0, t) \}^2 = \hat{P}_{00}^2(0, t).$$

Thus, the variance estimator for $\hat{P}_{00}(0, t)$ is:

$$\hat{\mathbb{V}}\{\hat{P}_{00}(0, t)\} = \hat{P}_{00}^2(0, t) \int_0^t \frac{dN_{01}(u)}{Y_0^2(u)},$$

in agreement with the standard estimator. ♠

1.3.2 Competing Risks Model

Example 1.5. For the J -state competing risks model, the variance of the survival function $\hat{P}_{00}(0, t)$ is:

$$\hat{V}\{\hat{P}_{00}(0, t)\} = \hat{P}_{00}^2(0, t) \int_0^t \frac{dN_{0\bullet}(u)}{Y_0^2(u)}.$$

To find the variance of $\hat{P}_{0j}$, note that $dN_{ki}(u) = 0$ for $k \neq 0$:

$$\hat{V}\{\hat{P}_{0j}(0, t)\} = \sum_{i=1}^J \int_0^t \hat{P}_{00}^2(0, u) \{\hat{P}_{ij}(u, t) - \hat{P}_{0j}(u, t)\}^2 \cdot \frac{dN_{0i}(u)}{Y_0^2(u)}.$$

$\hat{P}_{ij} = 0$ for $i \neq j$ and $\hat{P}_{jj} = 1$:

$$\begin{aligned} \hat{V}\{\hat{P}_{0j}(0, t)\} &= \sum_{i \neq j} \int_0^t \hat{P}_{00}^2(0, u) \{\hat{P}_{0j}(u, t)\}^2 \cdot \frac{dN_{0i}(u)}{Y_0^2(u)} \\ &\quad + \int_0^t \hat{P}_{00}^2(0, u) \{1 - \hat{P}_{0j}(u, t)\}^2 \cdot \frac{dN_{0j}(u)}{Y_0^2(u)}. \end{aligned}$$

Adding and subtracting $\int_0^t \hat{P}_{00}^2(0, u) \hat{P}_{0j}^2(u, t) \frac{dN_{0j}(u)}{Y_0^2(u)}$:

$$\begin{aligned} \hat{V}\{\hat{P}_{0j}(0, t)\} &= \int_0^t \hat{P}_{00}^2(0, u) \hat{P}_{0j}^2(u, t) \cdot \frac{dN_{0\bullet}(u)}{Y_0^2(u)} \\ &\quad + \int_0^t \hat{P}_{00}^2(0, u) \{1 - \hat{P}_{0j}(u, t)\}^2 \cdot \frac{dN_{0j}(u)}{Y_0^2(u)} \\ &\quad - \int_0^t \hat{P}_{00}^2(0, u) \hat{P}_{0j}^2(u, t) \cdot \frac{dN_{0j}(u)}{Y_0^2(u)}. \end{aligned}$$

Expanding the binomial gives:

$$\begin{aligned} \hat{V}\{\hat{P}_{0j}(0, t)\} &= \int_0^t \hat{P}_{00}^2(0, u) \hat{P}_{0j}^2(u, t) \cdot \frac{dN_{0\bullet}(u)}{Y_0^2(u)} \\ &\quad + \int_0^t \hat{P}_{00}^2(0, u) \{1 - 2\hat{P}_{0j}(u, t)\} \cdot \frac{dN_{0j}(u)}{Y_0^2(u)}. \end{aligned}$$



1.3.3 Illness-Death Model

Example 1.6. For the illness-death model without recovery, $\hat{P}_{00}(0, t)$ and $\hat{P}_{11}(0, t)$ are the standard Kaplan-Meier estimators, and their variance estimators are given by:

$$\hat{V}\{\hat{P}_{00}(0, t)\} = \hat{P}_{00}^2(0, t) \int_0^t \frac{dN_{0\bullet}(u)}{Y_0^2(u)}, \quad \hat{V}\{\hat{P}_{11}(0, t)\} = \hat{P}_{11}^2(0, t) \int_0^t \frac{dN_{12}(u)}{Y_1^2(u)}.$$

For the variance of $\hat{P}_{01}(0, t)$, the non-zero terms are $(i, k) \in \{(1, 0), (2, 0), (2, 1)\}$:

$$\begin{aligned}\hat{\mathbb{V}}\{\hat{P}_{01}(0, t)\} = & \int_0^t \hat{P}_{00}^2(0, u) \{\hat{P}_{11}(u, t) - \hat{P}_{01}(u, t)\}^2 \cdot \frac{dN_{01}(u)}{Y_0^2(u)} \\ & + \int_0^t \hat{P}_{00}^2(0, u) \{0 - \hat{P}_{01}(u, t)\}^2 \cdot \frac{dN_{02}(u)}{Y_0^2(u)} \\ & + \int_0^t \hat{P}_{01}^2(0, u) \{0 - \hat{P}_{11}(u, t)\}^2 \cdot \frac{dN_{12}(u)}{Y_1^2(u)}\end{aligned}$$



1.4 Computation

- The **Nelson-Aalen** estimator $\hat{\mathbf{A}}(t)$ of the cumulative hazard matrix $\mathbf{A}(t)$ for multi-state models is implemented by the `mvna` function in the `mvna` package.
- The **Aalen-Johansen** estimator $\hat{\mathbf{P}}(s, t)$ of the transition matrix $\mathbf{P}(s, t)$ for multi-state models is implemented by the `etm` function in the `etm` package.
- See [2] for examples using the `mvna` and `etm`.

References

- [1] PK Andersen et al. *Statistical Models Based on Counting Processes*. 2nd. Springer-Verlag, 1997.
- [2] J Beyersmann, A Allignol, and M Schumacher. *Competing Risks and Multistate Models with R*. Springer Science+Business Media, 2012.